

Lifting the Exponent Lemma

Problem 1. Let n be an odd positive integer. Prove that $((n-1)^n + 1)^2$ divides $n(n-1)^{(n-1)^n+1} + n$.

Problem 2. Given a prime p and a positive integer a , let q be a prime divisor of $\frac{a^{p^3} - 1}{a^{p^2} - 1}$ and $q \neq p$. Prove that $q \equiv 1 \pmod{p^3}$.

Problem 3. Let a and b be distinct real numbers such that $a^n - b^n$ is an integer for every positive integer n . Prove that a and b are integers.

Problem 4. Find all nonnegative integers m, n and all prime numbers p satisfying

$$2p^m - n^p = 1.$$

Problem 5. Find all the pairs of positive integers (x, p) such that p is a prime and x^{p-1} is a divisor of $(p-1)^x + 1$.

Problem 6. Solve the equation $(a^5 + b)(b^5 + a) = 2^c$ in positive integers.

Problem 7. Find all nonnegative integers m and n , such that $(2^n - 1) \cdot (3^n - 1) = m^2$.